HOMEWORK #3

#1 (a) $P = 100\ lbf/in^2$ $\quad u = 500\ \frac{Btu}{lb}$ $\quad T = ?\quad v = ?\quad h = ?$

from tables; $T = 327.86°F$ $\quad v_f = 0.01774\ ft^3/lb$

$u = u_f + x(u_g - u_f)$ $\quad v_g = 4.434\ ft^3/lb$

$u_f = 298.3\ Btu/lb$ $\quad u_g = 1105.8\ \frac{Btu}{lb}$

$500 = 298.3 + x(1105.8 - 298.3)$

$x = \frac{500 - 298.3}{1105.8 - 298.3} = 0.2498$

$v = v_f + x(v_g - v_f)$

$v = 0.01774 + 0.2498(4.434 - 0.01774)$

$v = 1.121\ ft^3/lb$

$h = h_f + x(h_g - h_f)$ $\quad h_f = 298.6\ Btu/lb$

$h = 298.6 + 0.2498(889.2 - 298.6)$ $\quad h_g = 889.2\ Btu/lb$

$h = 446.13\ Btu/lb$

(b) Ammonia; $T = 40°F$; $x = 0.5$ $\quad P = ?\quad v = ?\quad u = ?$

from tables; $P = 73.359\ lbf/in^2$

$v_f = 0.02533\ ft^3/lb$ $\quad v_g = 3.9664\ ft^3/lb$

$v = v_f + x(v_g - v_f)$; $v = 0.02533 + 0.5(3.9664 - 0.02533)$

$v = 1.996\ ft^3/lb \simeq 2\ ft^3/lb$

$u = u_f + x(u_g - u_f)$; $\quad u_f = 85.98\ \frac{Btu}{lb}$ $\quad u_g = 568.42\ \frac{Btu}{lb}$

$u = 85.98 + 0.5(568.42 - 85.98) = 327.2\ \frac{Btu}{lb}$

ⓒ Water ; $P = 80$ bar $v = 0.04034 \frac{m^3}{kg}$ $T = ?$ $u = ?$ $h = ?$

from tables ; $T = 480°C$ $u = 3025.7 \frac{kJ}{kg}$ $h = 3348.4 \frac{kJ}{kg}$

ⓓ Refrigerant 134a ; $T = -20°C$ $h = 235.31 \frac{kJ}{kg}$ $P = ?$ $v = ?$ $u = ?$

from tables ; $P = 1.3299$ bar $h_g = 235.31 \frac{kJ}{kg} = h$

$u = u_g = 215.84 \frac{kJ}{kg}$; $v = v_g = 0.1464 \, m^3/kg$

#2 State 1 : Saturated vapor

$T = 180°C$

from tables

initial pressure $P_1 = 10.02$ bar

$10.02 \text{ bar} \times \frac{0.1 MPa}{1 \text{ bar}} = 1.002 MPa$

$v_1 = 0.1941 \ m^3/kg$

State 2

$T = 100°C$ $\quad v_2 = 0.1941 \ \frac{m^3}{kg}$

$x = \frac{v - v_f}{v_g - v_f}$ $\quad v_f = 1.0435 \times 10^{-3} \ \frac{m^3}{kg}$

$v_g = 1.673 \ m^3/kg$

$x = \frac{0.1941 - 1.0435 \times 10^{-3}}{1.673 - 1.0435 \times 10^{-3}} = \frac{0.1930565}{1.6719565}$

$= 0.115$

#3 $m = 2\,kg$

State 1 : saturated vapor

(a) $P_1 = 300\,KPa = 3\,bar$

from tables

$T_1 = 133.6°C$

State 2

$P_2 = 300\,KPa = 0.3 \times 10^3\,KPa = 0.3\,MPa$

$V_2 = 2.064\,m^3$

$v_2 = \frac{V}{m} = \frac{2.064\,m^3}{2\,kg} = 1.032\,m^3/kg$

(b)

v_1 (Specific volume in state 1) $\Rightarrow$ $\frac{P_1 - P_2}{P_1 - P_x} = \frac{v_1 - v_2}{v_1 - v_x}$

$\frac{2.701 - 3.613}{2.701 - 3.0} = \frac{0.6685 - 0.5089}{0.6685 - v_x}$

$\frac{-0.912}{-0.299} = \frac{0.1596}{0.6685 - v_x} \Longrightarrow 3.0502 = \frac{0.1596}{0.6685 - v_x}$

$2.039 - 3.0502 v_x = 0.1596$; $3.0502\, v_x = 1.8794$

$v_x = 0.6162\,m^3/kg$ (initial specific volume) v_1

since $v_2 > v_x$ State 2 is superheated

from tables

Final temperature 400°C

$V_1 = v_1 \times m = 0.6162\,m^3/kg \times 2\,kg = 1.2324$

(c) $W = \int P\,dv$; $\Delta V = (V_2 - V_1)$

$W = 300\,KPa \times (2.064 - 1.2324)\,m^3 = 300 \times 10^3\,Pa \times 0.8316\,m^3$

$= 249480\,J \times \frac{1\,KJ}{1000\,J} = 249.5\,KJ$

refrigerant 134a

mass = 1 Kg

4 State 1;

$P_1 = 10 bar$ $T_1 = 60°C$

$V_1 = ?$

from table

$v_1 = 0.02301 \ m^3/Kg$

State 2

$P_2 = 2 bar$ $T_2 = 30°C$

$u_2 = 253.06 \ \frac{KJ}{Kg}$

Work = -20 KJ

$v_1 = \frac{V_1}{m}$; $V_1 = v_1 \times m = 0.02301 \ \frac{m^3}{Kg} \times 1 Kg = 0.02301 m^3$

$u_1 = 268.35 \ \frac{KJ}{Kg}$

$$\Delta u + \cancel{\Delta K.E} + \cancel{\Delta P.E} = Q + W \ ; \ \Delta u = Q + W$$

$u_2 - u_1 = Q + W$; $Q = (u_2 - u_1) - W$

$$Q = \left(253.06 \frac{KJ}{Kg} \times 1 Kg - 268.35 \frac{KJ}{Kg} \times 1 Kg\right) + 20 KJ = 4.71 KJ$$

#5 $m = 1.5\,kg$; Water

State 1 : Sat. liquid vapor

$T_1 = 70^\circ C$

from tables $P_1 = 0.3119\,bar$

$0.3119\,bar \times \frac{10^2\,KPa}{1\,bar} = 31.19\,KPa$

State 2 : sat. vapor

$T_2 = 120^\circ C$

$V_2 = 0.8919\,m^3/Kg$

$u_2 = 2529.3\,KJ/Kg$

(b) $Q = ?$

for state 1 $V_g = 5.042\,\frac{m^3}{Kg}$ $V_f = 0.0010228\,\frac{m^3}{Kg}$ $V_1 = 0.8919\,\frac{m^3}{Kg}$

$x = \frac{V - V_f}{V_g - V_f}$; $x = \frac{0.8919 - 0.0010228}{5.042 - 0.0010228} = 0.1767$

$u_f = 292.95\,\frac{KJ}{Kg}$ $u_g = 2469.6\,KJ/Kg$

$u_1 = u_f + x(u_g - u_f)$; $u_1 = 292.95 + 0.1767\,(2469.6 - 292.95)$

$u_1 = 677.56\,\frac{KJ}{Kg}$

$\Delta U + \cancel{\Delta K.E} + \cancel{\Delta P.E} = Q + \cancel{W}$; $\Delta U = Q$; $U_2 - U_1 = Q$

$(U_2 \times m) - (U_1 \times m) = Q$

$(2529.3\,\frac{KJ}{Kg} \times 1.5\,Kg) - (677.56\,\frac{KJ}{Kg} \times 1.5\,Kg) = 2777.61\,KJ = Q$

#6. Refrigerant Sat. liquid vapor

$P = 10\,bar$ $m = 25\,kg$ $x = 60\% = 0.6$

from tables; $v_f = 0.0008352\ \frac{m^3}{kg}$ $v_g = 0.0236\ m^3/kg$

$v_1 = v_f + x(v_g - v_f)$; $v_1 = 0.0008352 + 0.6(0.0236 - 0.0008352)$

$v_1 = 0.0145\ m^3/kg$;

$x = \frac{m_g}{m_g + m_f}$ $m_f = 25\,kg$

$0.6 = \frac{m_g}{m_g + 25}$; $0.6 m_g + 15 = m_g$; $0.4 m_g = 15 \Rightarrow m_g = 37.5\,kg$

Total mass = $37.5 + 25 = 62.5\,kg$

$V_1 = v_1 \times m_{total} = 0.0145\ \frac{m^3}{kg} \times 62.5\,kg = 0.90625\,m^3$

(b) $\frac{V_{sat.\ vapo}}{V_{total}} \times 100\%$; $V_{sat\ vapor} = v_g \times m_g = 0.0236 \times 37.5 = 0.885$

; $\frac{0.885}{0.90625} \times 100\% = 97.66\%$

#7 $m = 0.2g$ $M_{He} = 4.003 \frac{kg}{kmol}$

State 1

$T_1 = 27°C + 273.15 = 300.15°K$ $P_1 = 1.3 bar$

State 2 :

$T_2 = 17°c + 273.15 = 290.15°K$ $P = 0.8 bar$

ⓐ $\bar{R} = 8.314 \frac{KJ}{Kmol \cdot K}$; $R = \frac{\bar{R}}{M} = \frac{8.314 \, KJ}{\cancel{Kmol} \cdot K} \times \frac{1 \, \cancel{Kmol}}{4.003 \, Kg} = \frac{2.077 KJ}{K \cdot Kg}$

$\frac{2.077 \cancel{KJ}}{K \cdot \cancel{Kg}} \times \frac{1 \cancel{Kg}}{1000 g} \times \frac{1000 J}{1 \cancel{KJ}} = 2.077 \frac{J}{g \cdot K}$

ⓑ $V_i = ?$ $R = \frac{Pv}{T}$; $v_1 = \frac{RT}{P}$; $\bar{v} = \frac{v}{M}$

$v_1 = \frac{2.077 J}{g \cdot \cancel{K}} \times 300.15 \cancel{K} \times \frac{1}{1.3 \cancel{bar}} \times \frac{1 \cancel{bar}}{10^2 \cancel{KPa}} \times \frac{1 \cancel{KPa}}{1000 Pa} = 0.004795$

$v_1 = \frac{V}{m}$; $V = v_1 \times m = 0.004795 \times 0.2g = 0.9591 \times 10^{-3} m^3$

$0.9591 \times 10^{-3} \cancel{m^3} \times \left(\frac{100 cm}{1 \cancel{m}}\right)^{\cancel{3}} = 959.1 \, cm^3$

ⓒ $v_2 = \frac{RT}{P}$; $\frac{2.077 J}{g \cdot \cancel{K}} \times 290.15 \cancel{K} \times \frac{1}{0.8 \cancel{bar}} \times \frac{1 \cancel{bar}}{10^2 \cancel{KPa}} \times \frac{1 \cancel{KPa}}{1000 Pa} = 0.007533$

$V_2 = v_2 \times m = 0.007533 \times 0.2 = 1.5066 \times 10^{-3} m^3$

$1.5066 \times 10^{-3} \cancel{m^3} \times \left(\frac{100 cm}{1 \cancel{m}}\right)^{\cancel{3}} = 1506.6 \, cm^3$

#8. state 1 CO_2 $V_1 = 0.05\,m^3$ | state 2 $V_2 = 0.05\,m^3$

$T_1 = 25°C + 273.15 = 298.15\,K$ $P_1 = 2.75\,bar$ | $T_2 = 257°C + 273.15 = 530.15\,K$

$Q = -2.75\,KJ$ $m = ?$ $W = ?$ $M = 44.01\,kg/Kmol$

$$R = \frac{\bar{R}}{M} = \frac{8.314\,\cancel{K}J}{\cancel{Kmol}\cdot K} \times \frac{1\,\cancel{Kmol}}{44.01\,Kg} = 0.1889\,\frac{J}{g\cdot K}$$

ⓐ $PV = mRT$; $m = PV/RT$

$$2.75\,\cancel{bar} \times \frac{10^2\,K\cancel{Pa}}{1\,\cancel{bar}} \times 0.05\,\cancel{m^3} \times \frac{1\,g\cdot\cancel{K}}{0.1889\,\cancel{J}} \times \frac{1}{298.15\,\cancel{K}} = 0.244\,Kg$$

ⓑ $\cancel{\Delta K.E} + \cancel{\Delta P.E} + \Delta U = Q + W$; $\Delta U = Q + W$; $W = \Delta U - Q$

$u_1 = 6885\,\frac{KJ}{Kmol}$ $u_2 = 14622\,KJ/Kmol$ $n = \frac{m}{M}$

$$n = \frac{0.244\,Kg}{44.01\,\frac{Kg}{Kmol}} = 0.005544\,Kmol$$

$W = n(u_2 - u_1) - Q$; $0.005544\,Kmol\,(14622 - 6885)\frac{KJ}{Kmol} + 2.75\,KJ$

$W = 45.83\,KJ$